Antonio de Leon

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Professional Summary

Statistics Ph.D. candidate and applied quantitative researcher with experience in probabilistic forecasting, statistical learning, Bayesian computation, and reproducible model evaluation. Builds analytical workflows in Python, R, SQL, and C++ across financial, sensor, image, and environmental data, from data construction through diagnostics, software, and communication.

Education

Ph.D. in Statistics, University of California, Santa Cruz Expected Fall 2026
GPA: 3.82/4.00; candidacy completed. Research: probabilistic forecasting, model evaluation, and statistical software.

M.Sc. in Economic Theory, ITAM Jan. 2019–Dec. 2020
GPA: 96.7/100.

B.Sc. in Applied Mathematics, ITAM Jan. 2014–Dec. 2018
GPA: 97.1/100.

Relevant coursework and training: probability, statistical inference, regression, statistical learning, Bayesian inference, time series, stochastic processes, decision theory, causal inference, and impact evaluation.

Technical and Quantitative Skills

Programming: Python, R, SQL, C++, Bash, Git, LaTeX.
Statistical and machine-learning methods: probability, regression, Bayesian inference, time-series and state-space models, statistical learning, Monte Carlo methods, variational inference, experimental design, causal inference and impact-evaluation training, model diagnostics, calibration, and proper scoring.
Systems and tools: Linux, AWS (EC2, EMR, S3), Stan, Rcpp, CRAN, GitHub Actions, Tableau, DBeaver.
Languages: Spanish, English, German.

Professional Experience

Delos Financial Technologies, San Rafael, CA (Remote) May 2025–Sep. 2025
Quantitative Researcher; Python, Git, Bash, C++, AWS, Stan

- Built Python model-evaluation workflows for residual diagnostics, goodness-of-fit analysis, cohort summaries, and backtest review across financial-modeling experiments.
- Implemented baseline and extended forecasting specifications, automated AWS experiment runs, and standardized versioned outputs for reproducible model comparison.

UCSC Statistics Department, Santa Cruz, CA Feb. 2024–Present
Computer Systems Coordinator; Linux, Bash, Python/R, Networking, User Support

- Administer Linux research infrastructure for faculty and graduate students, including software stacks, storage, upgrades, access, and incident response.
- Write automation scripts and documentation for onboarding, resource usage, and reproducible computing; coordinate with central IT on research infrastructure planning.

NeatLeaf Inc., Scotts Valley, CA May 2022–Sep. 2022
Data Analyst; Python, SQL, Git, Bash, Linux, C++, DBeaver

- Built Python and SQL pipelines for greenhouse robot and sensor telemetry used in operational monitoring and downstream analysis.
- Analyzed spatiotemporal microclimate anomalies and robot/sensor calibration logs to improve data-quality review and system reliability.

Banco de México (BANXICO), Mexico City, Mexico Aug. 2018–Dec. 2019
Data Analyst; Python, R, SQL, Git

- Built Python/R/SQL pipelines for banknote image and operational datasets used in monitoring, anomaly analysis, and model development.
- Developed anomaly-classification diagnostics and financial-volume forecasting prototypes for operational planning and review.

Selected Research and Software

exdqlm – CRAN R package v1.1.2 2026
Software article submitted to the Journal of Statistical Software

- Co-developed a public package for Bayesian dynamic and static quantile models, including MCMC and variational inference, forecasting, diagnostics, tests, documentation, and reproducible examples.
- Added evaluation utilities and selected C++ paths, with R reference implementations for validation and portability.

Bayesian Quantile Deep Echo State Networks for Nonlinear Time Series 2024–Present

arXiv preprint on nonlinear time-series forecasting

- Developed probabilistic time-series models using fixed nonlinear recurrent features, regularization, MCMC, and variational inference.
- Designed simulation and rolling held-out studies, including a selected-panel electricity-price forecasting comparison with 42 region/fold evaluations.

Bayesian Quantile-Based Correction and Synthesis of Hydrologic Products 2021–Present

arXiv preprint; manuscript submitted to Environmetrics

- Integrated USGS observations, retrospective products, ensemble forecasts, and covariates from multiple agencies into origin-specific, version-consistent datasets.
- Evaluated probabilistic forecasts across five origins and seven fitted quantile levels using CRPS and quantile loss, keeping post-origin observations reserved for verification.

Selected Publications

De Leon, A., Barata, R., Prado, R., & Sansó, B.
exdqlm: An R Package for Estimation and Analysis of Flexible Dynamic Quantile Linear Models.
Submitted to the *Journal of Statistical Software*; CRAN R package v1.1.2; arXiv:2607.22760.

De Leon, A., Prado, R., & Sansó, B.
Bayesian Quantile Deep Echo State Networks for Nonlinear Time Series.
arXiv preprint on fixed deep-reservoir features, Bayesian quantile regression, MCMC, variational inference, and retrospective forecast comparisons; arXiv:2609.17579.

De Leon, A., Prado, R., & Sansó, B.
Mean-Tilted Intervals: A Generalized-Bayes Approach to Fixed-Content and Tolerance Intervals.
arXiv preprint on MTI fixed-content and tolerance-interval targets, generalized-Bayes computation, and calibrated tolerance actions; arXiv:2607.26098.

De Leon, A., Prado, R., & Sansó, B.
Mean-Tilted Intervals: Regression and Dynamic Models.
Submitted manuscript on MTI regression and dynamic state-space extensions; public identifier pending.

De Leon, A., Prado, R., & Sansó, B.
Bayesian Quantile-Based Correction and Synthesis of Hydrologic Products.
arXiv preprint; manuscript submitted to *Environmetrics*; arXiv:2608.11222.

De Leon, A., & Lobato, I. N.
Evidence of non-fundamentalness in OECD capital stocks.
Empirical Economics 67, 761–772 (2024).

Awards, Teaching, and Service

Awards: Best Poster Prize, ISBA 2026 World Meeting; EnviBayes Student Paper Competition Winner, 2026; ISBA 2026 Junior Researcher Travel Award; UC Regents Fellowship, UCSC Statistics, 2021; Baillères Fellowship, ITAM, 2014–2020.

Teaching and mentoring: Instructor of record / Graduate Student Instructor for *Probability and Statistics for Engineers* (CSE 107), Fall 2026, approximately 150 students; Graduate Student Instructor for *Data Visualization*; Teaching Assistant for *Probability Theory*, *Classical and Bayesian Inference*, *Statistics*, and related courses; ASA DataFest mentor for student teams in modeling, visualization, and presentation.

Service and training: TA Wiki maintainer and workshop contributor for Git, GitHub, collaborative documentation, and departmental teaching resources; contributed to an anonymous, IRB-exempt TA Wiki survey and follow-up resource improvements.